Risk Factors for Diabetes Among Adults Aged 40+

A Combined Logistic Regression and Random Forest Analysis Using NHANES Data

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Abstract

Diabetes disproportionately affects U.S. adults aged 40 years and older, yet the combined influence of socioeconomic, behavioral, and psychosocial determinants remains understudied. Using data from 2,873 adults in the 2009–2012 NHANES cycles, this study examined (1) how these factors predict diabetes risk and (2) whether variable importance rankings differ between logistic regression and a Random Forest model. After extensive data cleaning, transformation, and KNN imputation, we fitted a multivariable logistic regression model and trained a Random Forest classifier with cross-validation and downsampling to address class imbalance. Logistic regression identified age, race, low income, home rental status, short sleep duration, sleep trouble, computer use, and depressive symptoms as significantly associated with diabetes. Random Forest variable importance rankings showed strong concordance with logistic regression, particularly for age, race, poverty, gender, and sedentary behaviors, while highlighting potential nonlinear effects for marital status, physical activity, and smoking. Together, findings underscore the multifactorial nature of diabetes risk and the value of integrating statistical and machine-learning approaches to identify both linear and complex determinants of disease among aging adults.

Introduction

Diabetes mellitus remains one of the most pressing chronic diseases affecting U.S. adults, with prevalence increasing substantially with age. According to the CDC's National Diabetes Statistics Report, 18.9% of adults aged 45-64 and 29.2% of adults aged 65 years and older have diabetes, reflecting an urgent need to understand the factors that place middle-aged and older adults at elevated risk (Centers for Disease Control and Prevention, 2024). Type 2 diabetes, which accounts for the vast majority of cases, develops through a multifactorial interplay of behavioral, psychosocial, metabolic, and socioeconomic determinants. As the population ages, identifying modifiable risk factors that influence diabetes onset among adults aged 40 and

older is critical for guiding effective clinical screening, public health planning, and targeted intervention strategies.

A substantial literature has examined the role of socioeconomic status, lifestyle factors, and psychosocial stressors in shaping diabetes risk. Socioeconomic status, for example, has consistently been recognized as having one of the strongest associations with diabetes. Analysis using national socioeconomic survey data demonstrated that gender, age, race, education, marital status, occupation, income, and debt levels were significantly associated with diabetes prevalence (Suwannaphant et al., 2017). Similarly, a 2023 study using NHANES data found a negative nonlinear association between income-to-poverty ratio and diabetes risk, independent of demographic and behavioral confounders, indicating that participants with lower socioeconomic status had a higher risk of diabetes (Liu et al., 2023). Behavioral factors, including physical inactivity, sleep quantity or quality, alcohol consumption patterns, and smoking, also play a well-established role in diabetes development (Ismail et al., 2021). Sleep duration, in particular, has been shown to exhibit a U-shaped relationship with diabetes risk: both short sleep ($<$6 hours) and long sleep ($>$9 hours) are associated with elevated odds compared to adequate sleep (Cappuccio et al., 2010; Shan et al., 2015). Similarly, while moderate alcohol consumption has been hypothesized to confer metabolic benefits in some populations, observational studies face challenges in disentangling causal effects from confounding by socioeconomic status and lifestyle patterns. Recent research has also highlighted the role of psychosocial determinants, demonstrating that symptoms of depression and anxiety contribute meaningfully to diabetes risk, potentially through both behavioral pathways (e.g., poor adherence to healthy lifestyles) and physiological mechanisms (e.g., chronic stress-induced metabolic dysregulation) (Naicker et al., 2018). Despite extensive evidence, relatively few studies simultaneously examine socioeconomic, behavioral, and psychosocial factors within a single representative dataset for U.S. adults aged 40 and older.

Furthermore, most prior studies have relied exclusively on traditional regression-based approaches, which, while interpretable, may overlook complex interactions and nonlinear relationships among risk factors. Machine learning methods, such as random forests, offer complementary advantages by capturing intricate patterns without assuming linearity or requiring explicit specification of interaction terms. Comparing the variable importance rankings from machine learning models with the statistically significant predictors identified through logistic regression can provide deeper insights into which factors most reliably predict diabetes risk across different analytical frameworks.

The present study addresses these gaps by leveraging data from the National Health and Nutrition Examination Survey (NHANES) to investigate two primary research questions: (1) Among U.S. adults aged 40 years and older, how do socioeconomic, behavioral, and psychosocial factors influence the risk of diabetes? (2) How do the variable importance rankings from a random forest model compare with the statistically significant predictors from logistic regression? By integrating classical statistical inference with modern machine learning, this study aims to provide a comprehensive, multidimensional assessment of diabetes risk factors, with implications for targeted prevention and intervention strategies in an aging population.

Methods

Data Source and Study Population

We used data from the National Health and Nutrition Examination Survey (NHANES) R package, which provides a streamlined dataset of 10,000 observations from the 2009–2010 and 2011–2012 survey cycles. NHANES integrates standardized clinical measurements with detailed demographic, socioeconomic, behavioral, and psychosocial survey data, making it well suited for modeling diabetes risk factors (Supplementary Table 1).

Because our objective was to assess diabetes risk factors and compare variable importance across modeling approaches, we used the NHANES package rather than the survey-weighted NHANESraw data, avoiding additional complexity from sampling weights. Analyses were restricted to adults aged 40 years. After removing duplicate records and excluding pregnant women, the final analytic sample comprised 2,873 participants.

Data Cleaning and Preprocessing

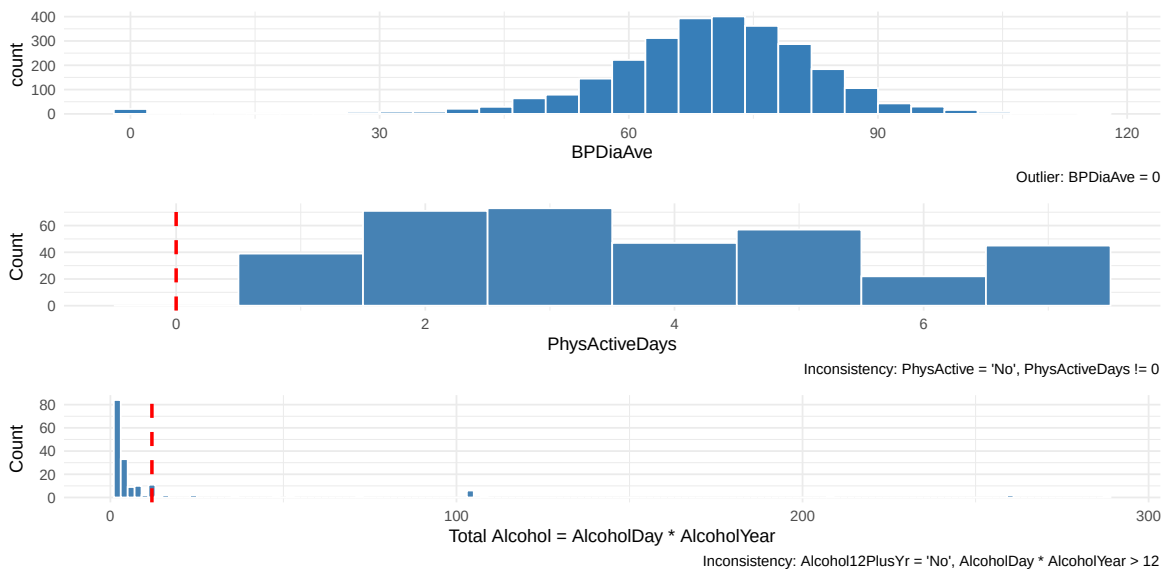


Figure 1: Data Quality: Detecting Outliers and Inconsistencies

Figure 1 identified several data quality issues requiring correction prior to analysis. Diastolic blood pressure (BPDiaAve) contained biologically implausible zero values, which were recoded as missing. Logical inconsistencies were observed in physical activity reporting: participants reporting no moderate or vigorous activity (PhysActive = “No”) but positive PhysActiveDays

values; these were corrected by setting PhysActiveDays to zero. Alcohol variables also showed inconsistencies, whereby participants reporting fewer than 12 drinks in any one year (Alcohol12PlusYr = "No") had AlcoholDay × AlcoholYear values exceeding 12; these values were recoded as missing.

Missingness varied substantially across variables (Supplementary Table 2). To retain all participants, we applied k-nearest neighbors (KNN) imputation (k = 5). KNN imputes missing values based on similarity in observed covariates, preserving multivariable relationships more effectively than single-value imputation. Although KNN assumes missingness at random and is sensitive to the choice of k, it was preferred over mean/median imputation.

Variable Selection and Transformation

Variable selection was guided by theoretical relevance to diabetes, prior literature, and avoidance of post-treatment or mediator bias. We excluded: (1) variables not applicable to adults aged 40 years (child anthropometrics, pregnancy measures); (2) variables with limited diabetes relevance (testosterone, urinary flow, marijuana use, sexual behavior); (3) physiological mediators on the causal pathway (BMI, blood pressure, cholesterol); (4) post-treatment health indicators to prevent collider bias; and (5) redundant variables. This process yielded a parsimonious set of socioeconomic, behavioral, and psychosocial predictors.

Retained variables (Supplementary Figure 1) were harmonized and recoded for interpretability. Based on prior evidence of nonlinear associations with diabetes (Liu et al., 2023), poverty (IPR) was categorized as Low (1.0), Middle (1.0–4.0), or High (4.0). Education and marital status were collapsed into three-category measures. Sleep duration was categorized as Short (<7 h), Adequate (7–9 h), or Long (>9 h) to capture its U-shaped relationship with diabetes (Cappuccio et al., 2010; Shan et al., 2015). Smoking status was derived as Never, Former, or Current, and alcohol consumption was categorized using annual intake (AlcoholDay × AlcoholYear) and categorized as Non-drinker, Light–Moderate (<730 drinks/year), or Heavy (730 drinks/year). Ordinal sedentary behavior variables were converted to numeric midpoints, and depressive symptom indicators were dichotomized to reflect clinically meaningful symptom presence.

Exploratory Data Analysis

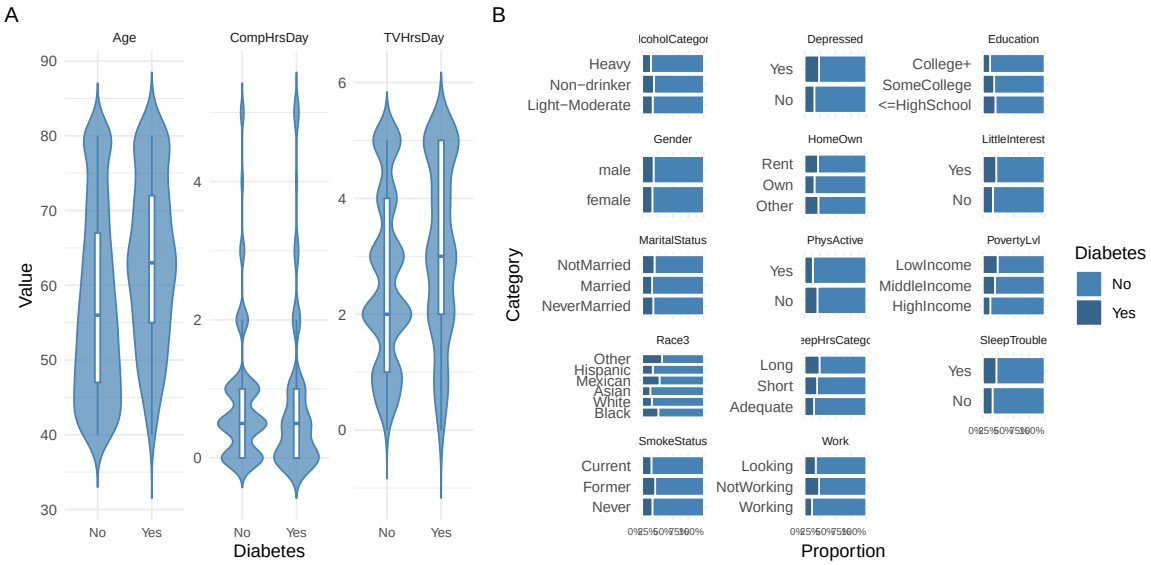


Figure 2: Numeric (Left) and Categorical Variables (Right) by Diabetes Risk

Following variable transformation, exploratory data analysis was conducted to examine variable distributions (Figure 2). Participants with diabetes were older on average, while sedentary behavior measures showed substantial overlap between outcome groups. Higher diabetes prevalence was observed among participants with lower income, lower education, sleep trouble, physical inactivity, and former smoking. No problematic multicollinearity was detected among retained predictors (all pairwise correlations $r < 0.7$; Supplementary Figure 2).

Statistical Analysis

We employed two complementary approaches to evaluate diabetes risk factors.

Logistic Regression

We fitted a multivariable logistic regression model with diabetes status (yes/no) as the outcome and all retained predictors as covariates. The model was specified as:

$$\log\left(\frac{P(\text{Diabetes} = \text{Yes})}{1 - P(\text{Diabetes} = \text{Yes})}\right) = \beta_0 + \beta_1 \text{Gender}_{\text{male}} + \beta_2 \text{Age} + \beta_3 \text{Race}_{\text{White}} + \beta_4 \text{Race}_{\text{Asian}} + \beta_5 \text{Race}_{\text{Mexican}}$$
$$+ \beta_6 \text{Race}_{\text{Hispanic}} + \beta_7 \text{Race}_{\text{Other}} + \beta_8 \text{Education}_{\text{SomeCollege}} + \beta_9 \text{Education}_{\text{College+}} + \beta_{10} \text{MaritalStatus}_{\text{Married}}$$
$$+ \beta_{11} \text{MaritalStatus}_{\text{NotMarried}} + \beta_{12} \text{Poverty}_{\text{MiddleIncome}} + \beta_{13} \text{Poverty}_{\text{LowIncome}} + \beta_{14} \text{Work}_{\text{NotWorking}}$$
$$+ \beta_{15} \text{Work}_{\text{Looking}} + \beta_{16} \text{HomeOwn}_{\text{Rent}} + \beta_{17} \text{HomeOwn}_{\text{Other}} + \beta_{18} \text{Sleep}_{\text{Short}} + \beta_{19} \text{Sleep}_{\text{Long}}$$
$$+ \beta_{20} \text{SleepTrouble}_{\text{Yes}} + \beta_{21} \text{PhysActive}_{\text{Yes}} + \beta_{22} \text{TVHrsDay} + \beta_{23} \text{CompHrsDay} + \beta_{24} \text{Smoke}_{\text{Former}}$$
$$+ \beta_{25} \text{Smoke}_{\text{Current}} + \beta_{26} \text{Alcohol}_{\text{NonDrinker}} + \beta_{27} \text{Alcohol}_{\text{Heavy}} + \beta_{28} \text{LittleInterest}_{\text{Yes}} + \beta_{29} \text{Depressed}_{\text{Yes}}$$

Odds ratios (ORs) with 95% confidence intervals were reported, enabling identification of statistically significant predictors and quantification of their associations with diabetes risk. Model diagnostics included variance inflation factors and standard diagnostic plots.

Random Forest

A Random Forest classifier was trained using the same predictors. The model was trained with five-fold cross-validation using the caret framework, with tuning of the mtry parameter and downsampling to address class imbalance. Performance was evaluated on an independent test set using ROC–AUC, sensitivity, specificity, and balanced accuracy. Variable importance was assessed using permutation-based importance, defined as the decrease in ROC–AUC following predictor permutation.

Comparison

We compared the top 15 predictors from Random Forest (ranked by permutation-based importance) with the top 15 predictors from logistic regression (ranked by |z-value|). Concordance and divergence between methods were used to assess robustness and potential nonlinear effects.

Results

Our final sample comprised 2,873 adults aged 40 years, of whom 502 (17.5%) had diabetes.

Logistic Regression Results

Table 1 presents estimated coefficients, odds ratios (ORs), and 95% confidence intervals (CIs) from the multivariable logistic regression model. Model diagnostics indicated no major violations: all variance inflation factors (VIFs) were below 5 (Supplementary Table 3), and diagnostic plots suggested adequate model fit (Supplementary Figure 3).

Age showed the strongest association with diabetes, with each additional year associated with a 4% increase in odds. Significant racial and ethnic disparities were observed: compared with Black participants, White, Asian, and Hispanic participants had lower odds of diabetes. Male participants had 33% higher odds than females.

Socioeconomic factors. poverty (IPR) demonstrated a graded association: low-income participants had 49% higher odds compared with high-income participants, while middle-income showed a marginal association (OR = 1.25, p = 0.087). Renting was associated with 41% higher odds compared with homeownership. Education, marital status, and employment were not significant.

Behavioral factors. Short sleep duration (<7 hours) was associated with higher diabetes odds, as was self-reported sleep trouble. Computer use showed a dose-response relationship (OR = 1.13 per hour, p = 0.010), while TV viewing showed borderline significance (OR = 1.07, p = 0.061). Physical activity trended toward protection. Current smoking was inversely associated with diabetes, while former smoking showed no association. Alcohol consumption was not significant.

Psychosocial factors. Depressive symptoms were associated with 43% higher diabetes odds, while little interest in activities was not significant.

Table 1: Logistic Regression Results for Diabetes Risk Factors

Variable	Estimate	z value	p-value	OR (95% CI)	
(Intercept)	-4.303	-9.756	<0.001	0.01 (0.01–0.03)	***
Gendermale	0.284	2.516	0.012	1.33 (1.07–1.66)	*
Age	0.040	6.827	<0.001	1.04 (1.03–1.05)	***
Race3White	-0.793	-4.949	<0.001	0.45 (0.33–0.62)	***
Race3Asian	-0.733	-2.473	0.013	0.48 (0.26–0.84)	*
Race3Mexican	0.260	1.063	0.288	1.30 (0.80–2.09)	
Race3Hispanic	-0.605	-2.203	0.028	0.55 (0.31–0.93)	*
Race3Other	0.250	0.577	0.564	1.28 (0.53–2.93)	
Education Some College	0.125	0.987	0.323	1.13 (0.88–1.45)	
Education College+	-0.079	-0.506	0.613	0.92 (0.68–1.25)	
Marital Status Married	0.023	0.112	0.911	1.02 (0.69–1.54)	
Marital Status Not Married	-0.105	-0.499	0.618	0.90 (0.60–1.37)	
Poverty Lvl Middle Income	0.226	1.709	0.087	1.25 (0.97–1.63)	.

Variable	Estimate	z value	p-value	OR (95% CI)	
Poverty Lvl Low Income	0.399	2.124	0.034	1.49 (1.03–2.15)	*
Work Not Working	0.215	1.611	0.107	1.24 (0.95–1.61)	
Work Looking	0.128	0.378	0.706	1.14 (0.56–2.14)	
Home Own Rent	0.341	2.537	0.011	1.41 (1.08–1.83)	*
Home Own Other	0.286	0.894	0.372	1.33 (0.69–2.44)	
Sleep Hrs Category Short	0.231	2.082	0.037	1.26 (1.01–1.56)	*
Sleep Hrs Category Long	0.129	0.404	0.686	1.14 (0.59–2.08)	
Sleep Trouble Yes	0.307	2.605	0.009	1.36 (1.08–1.71)	**
Phys Active Yes	-0.220	-1.905	0.057	0.80 (0.64–1.01)	.
TVHrs Day	0.072	1.873	0.061	1.07 (1.00–1.16)	.
Comp Hrs Day	0.121	2.584	0.010	1.13 (1.03–1.24)	**
Smoke Status Former	0.065	0.514	0.607	1.07 (0.83–1.36)	
Smoke Status Current	-0.298	-1.999	0.046	0.74 (0.55–0.99)	*
Alcohol Category Non-drinker	-0.032	-0.257	0.797	0.97 (0.76–1.23)	
Alcohol Category Heavy	-0.106	-0.414	0.679	0.90 (0.53–1.46)	
Little Interest Yes	0.079	0.546	0.585	1.08 (0.81–1.43)	
Depressed Yes	0.356	2.410	0.016	1.43 (1.07–1.90)	*

Random Forest Results

A Random Forest classifier was trained using the same set of socioeconomic, behavioral, and psychosocial predictors as the logistic regression model. Model performance was evaluated on an independent held-out test set.

The Random Forest achieved an overall accuracy of 0.645 (95% CI: 0.604–0.684). Because the outcome was imbalanced, balanced accuracy is also reported and equaled 0.678. Sensitivity was 0.730, indicating that 73.0% of individuals with diabetes were correctly classified, while specificity was 0.627, indicating that 62.7% of non-diabetic individuals were correctly identified. The positive predictive value was 0.292, and the negative predictive value was 0.917. The prevalence of diabetes in the test set was 17.4%.

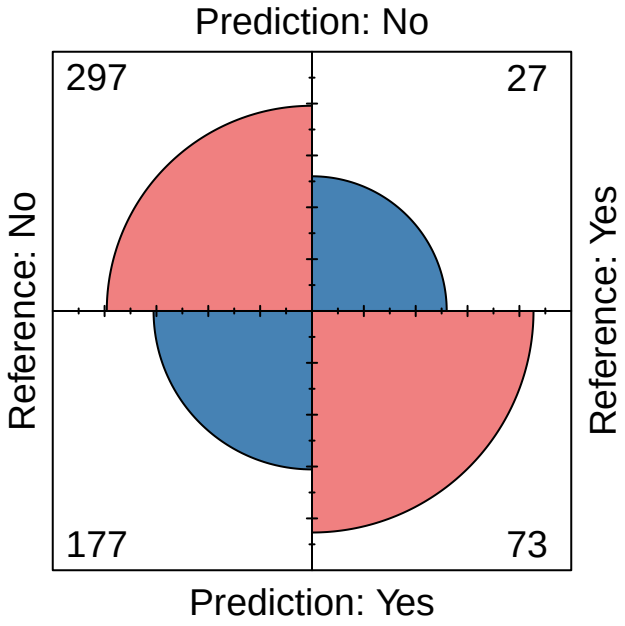


Figure 3: Confusion matrix for the Random Forest classifier on the test set.

The receiver operating characteristic (ROC) curve for the Random Forest model is shown in (Supplementary Figure 4). The area under the ROC curve (AUC) was 0.719, indicating good discrimination between individuals with and without diabetes across classification thresholds. Figure 3 presents the confusion matrix for the Random Forest classifier. Of 100 individuals predicted to have diabetes, 73 were true positives and 27 were false positives. Among those predicted not to have diabetes, 297 were true negatives and 177 were false negatives.

Comparison of Variable Importance

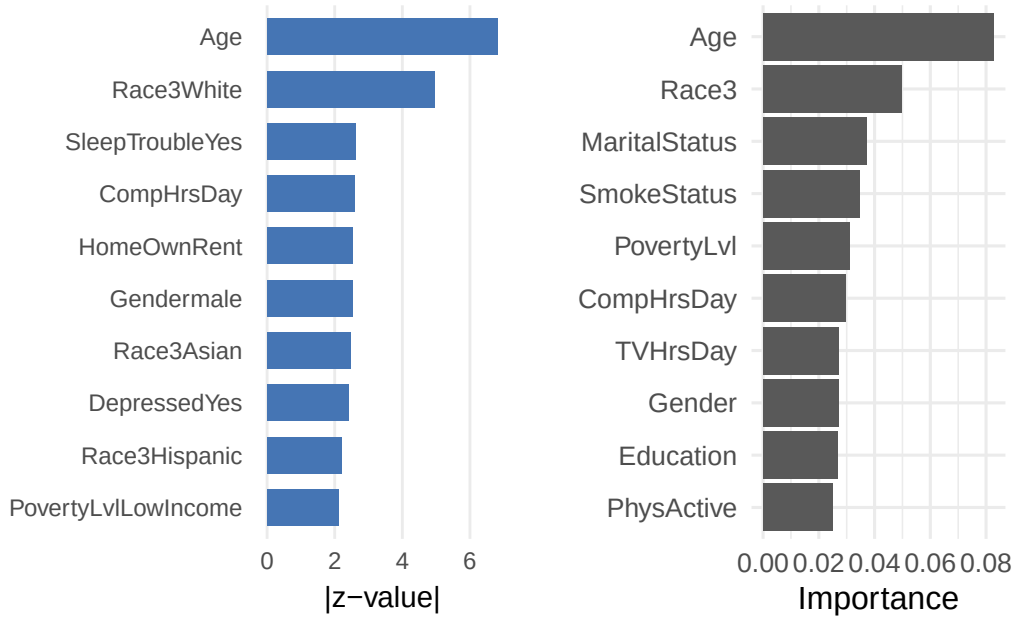


Figure 4: Comparison of Top 10 Variables Identified by Logistic Regression (Left) vs. Random Forest (Right)

Figure 4 compares top-10 predictors ranked by |z-value| (logistic regression) and permutation-based importance (Random Forest). Strong concordance was observed: age dominated both rankings, followed by race (White vs. Black). Five variables overlapped in both top-10 lists: age, race, poverty, gender, and computer use; all statistically significant in logistic regression.

Notable divergence emerged in mid-tier rankings. Sleep trouble and home ownership ranked 3rd and 5th in logistic regression but not in Random Forest's top 10. Conversely, marital status, smoking, TV hours, and physical activity ranked 3rd–6th in Random Forest despite weak or non-significant logistic regression associations.

Socioeconomic variables showed moderate importance: poverty appeared in both top-10 lists, while education ranked 7th in Random Forest but outside the logistic regression top 10. The substantial overlap strengthens confidence in these factors as robust diabetes predictors across parametric and non-parametric frameworks.

Discussion

This study examined socioeconomic, behavioral, and psychosocial determinants of diabetes among U.S. adults aged 40 years and older using complementary analytical approaches. Both logistic regression and Random Forest models consistently identified age, race, poverty level, and sedentary behavior as influential diabetes predictors. Notable divergence between methods emerged for marital status, smoking, and physical activity, suggesting potential nonlinear relationships warranting further investigation.

Demographic and socioeconomic factors dominated both models. Age showed the strongest association (OR = 1.04 per year, $p < 0.001$), consistent with established links between aging and insulin resistance. Substantial racial disparities persisted: Black participants exhibited higher diabetes odds than White, Asian, and Hispanic participants, corroborating prior NHANES analyses (Menke et al., 2015). Low income (OR = 1.49, $p = 0.034$) and renting versus owning a home (OR = 1.41, $p = 0.011$) were associated with higher diabetes odds, established socioeconomic pathways (Hill-Briggs et al., 2021). Education showed no significant association in logistic regression despite moderate Random Forest importance, suggesting nonlinear or context-dependent effects.

Behavioral factors emerged as modifiable risk factors. Both short sleep duration (<7 hours; OR = 1.26, $p = 0.037$) and sleep trouble (OR = 1.36, $p = 0.009$) associated with diabetes, corroborating meta-analytic evidence that insufficient or poor-quality sleep disrupts glucose metabolism (Cappuccio et al., 2010; Shan et al., 2015). Computer usage hours demonstrated a dose-response relationship (OR = 1.13 per hour, $p = 0.010$), while TV viewing showed borderline significance (OR = 1.07, $p = 0.061$). Interestingly, Random Forest assigned TV hours higher importance, possibly reflecting nonlinear thresholds or interactions not captured by the parametric model. Physical activity trended toward protection (OR = 0.80, $p = 0.057$) but did not reach significance, likely due to self-report limitations. Depression showed a significant positive association (OR = 1.43, $p = 0.016$), consistent with bidirectional links involving shared pathophysiology and behavioral pathways (Naicker et al., 2018).

Current smoking showed an inverse association (OR = 0.74, 95% CI: 0.55–0.99, $p = 0.046$), contradicting prospective evidence. This paradox likely reflects survival bias, older smokers with diabetes face elevated mortality, selectively removing them from the cross-sectional sample—and reverse causation, whereby individuals quit smoking after diagnosis. Current smokers with diabetes were younger than former smokers with diabetes (Supplementary Table 4), supporting mortality selection. Former smokers showed no protective association (OR = 1.07, $p = 0.607$). Our results should not be interpreted as evidence that smoking protects against diabetes.

Alcohol consumption showed no significant associations. Heavy drinking exhibited a non-significant inverse association (OR = 0.90, 95% CI: 0.53–1.46, $p = 0.679$), inconsistent with meta-analytic evidence of a U-shaped relationship (Knott et al., 2015). This null finding likely

reflects limited power (small sample), substantial underreporting, and potential survival bias (Supplementary Table 5).

Comparison of logistic regression and Random Forest variable importance revealed both concordances and differences. Five variables (age, race, poverty, gender, computer usage hours) appeared in both top 10 lists, validating robustness. Marital status and smoking ranked highly in Random Forest despite weak logistic regression associations, suggesting threshold effects or interactions. Conversely, sleep trouble and home ownership ranked highly in logistic regression but not Random Forest, possibly reflecting predominantly linear effects.

Limitations

Several limitations warrant consideration. The cross-sectional design precludes causal inference and is vulnerable to reverse causation (smoking, physical activity, depression). KNN imputation relied on the missing-at-random assumption, potentially violated for alcohol and smoking. Self-reported measures are prone to recall and social desirability bias. Computer usage hours, TV watching hours, and race data were completely absent in 2009-2010, introducing potential temporal bias. While Random Forest can detect interactions, our analysis did not explicitly identify specific interaction terms.

Implications and Future Directions

Our findings support multifaceted diabetes prevention strategies targeting modifiable risk factors—particularly sleep health, sedentary behavior reduction, and socioeconomic interventions—alongside continued efforts to address structural racism and health inequities. Future research should employ prospective designs to establish temporality, utilize objective measures (actigraphy for sleep/activity, biomarkers for stress), and explicitly model interactions identified by Random Forest to elucidate underlying mechanisms. Machine learning methods may also inform personalized risk stratification, identifying subgroups for targeted interventions.

Conclusion

Integrating logistic regression and random forest approaches, we identified age, race, poverty, sleep disturbances, and sedentary behavior as robust diabetes risk factors among U.S. adults aged 40 and older. Addressing modifiable risk factors, particularly sleep health and sedentary behavior, alongside upstream social determinants remains critical for diabetes prevention in this high-risk population.

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